SEEDLINE · MARKET AND VIABILITY

# SeedLine market and viability

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## DOCUMENT CONTROL

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| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/seedline | | |
| Source | docs/10-market.md @ 0a57933 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.1 | 2026-10-24 | Amish Chadha | First issue (kit 1.8 pipeline, kit 1.9.1 upgrade): segments, cited benchmarks with dates, cost per unit from the BOM, open route |

# SeedLine market and viability

SeedLine is for smallholders planting maize, beans, sorghum and groundnut on 0.2 to 2 ha, and later for market gardeners, who today plant by hoe, stick or jab planter or with a low-cost plate seeder that drops several seeds per cell. Its estimated parts cost (USD 224.10 for the base seeder) sits just above a low-cost plate seeder and well under a singulating roller seeder before rollers; the open design matters because the part that must match each crop, the plate, becomes a print file a cooperative or local maker can produce rather than a spare part bought from one maker. Prices are cited with the date they were checked; nothing here is a sales forecast. Method: STANDARDS section 23.

Table 1. Market segments.

| Segment | Target users and region | Competitive benchmark (cited price) | Estimated cost per unit (USD) | Value proposition |
|---|---|---|---|---|
| Smallholder field crops | Farmers on 0.2 to 2 ha, maize, beans, sorghum and groundnut on 0.75 m rows; Eastern and Southern Africa first (partner rule, SDL-DEC-001), South Asia and Central America later | Hoe planting, about 56 h/ha for one worker, and jab planters at about a third of that time (CSBE study, accessed 2026-10-24); jab planter prices to assess | 224.10 (base seeder, parts only, `bom/bom.csv`); labour and margin not included | About 7.1 h/ha at 0.142 ha/h (SDL-CAL-001), spacing set by travel, depth set by the opener; plates printed for local varieties |
| Market gardens and urban farms | Growers sowing beds of 3 to 6 rows with pelleted vegetable seed; North America and Europe as well as peri-urban farms elsewhere | Earthway 1001-B Precision Garden Seeder with six plates, USD 170.16 (DoMyOwn, accessed 2026-10-24; about USD 187 in UMN Extension, 2024); Jang JP-1 push seeder without rollers, USD 498.00 (Neversink Tools sale price, accessed 2026-10-24), rollers sold separately, about USD 25 each (UMN Extension, 2024) | 224.10 base; 266.10 with the marker and ratio kits | Single-seed metering with a printed plate per seed and spacing, between the plate seeder's price and the roller seeder's |
| Cooperatives, agro-dealers and seed companies | Rural towns with one desktop printer; dealers selling seed varieties | No printed-plate service known; stock plates come from the seeder's maker (Paperpot Co., accessed 2026-10-24) | Plate about USD 1.25 of filament (BOM line 6); printer time not costed | Print plates on demand for members' varieties; rent seeders |
| Extension, NGOs and research | Demonstration plots, conservation agriculture programmes, spacing trials | Hand planting and jab planters in programmes (FAO jab planter manual, accessed 2026-10-24) | 266.10 (prototype with kits) | A low-cost tool whose spacing is set by a file, for demonstrations and trials |

Prices are list or sale prices from one seller each on the date shown, before shipping and duties; they are benchmarks, not quotes. A replicated seeder's selling price, margin and labour are not estimated at TRL 3.

## Open route

SeedLine is published under CERN-OHL-S-2.0 (hardware) and MIT (software), so a workshop, cooperative, NGO or university can build, repair and adapt it without a licence fee, and changes to the design stay open. The intended route is a co-design partner already working with smallholder maize and bean farmers (first candidate: CIMMYT's small-scale mechanization work in Eastern and Southern Africa; nothing is agreed), local welders and makers building frames from the build plan (SDL-BLD-001), and cooperatives or agro-dealers printing plates with the plate generator in `cad/src/model.py` once bench tests confirm its cell sizes (SDL-DEC-001, decided 2026-10-02). Who prints plates in practice, and how plates reach villages without printers, are open questions for the partner (SDL-PRB-001).

## Sources


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